Assignment 5 Report

Rahul Vimalkanth
EE23B061

October 2024

1 Introduction

This project aims to optimize keyboard layouts with the goal of reducing the total finger travel distance when typing a given text. The motivation behind this is to enhance typing efficiency and ergonomics by analyzing and improving commonly used keyboard layouts, such as QWERTY and Dvorak. The optimization process is carried out using the Simulated Annealing algorithm, which explores different layout configurations to minimize the finger travel distance. This report details the methodology employed, including the analysis of initial and optimized layouts, the visualization of key press frequencies using heatmaps, and insights into the advantages of such optimizations.

2 Simulated Annealing

Simulated Annealing is an iterative optimization algorithm that seeks to find near-optimal solutions in complex solution spaces. The algorithm is inspired by the process of annealing in metallurgy, where materials are heated and then slowly cooled to remove defects. Simulated Annealing's primary feature is its ability to explore a wide range of solutions by accepting both better and occasionally worse solutions during the search, depending on the system's current temperature. This flexibility enables the algorithm to escape local minima, increasing the likelihood of finding a global optimum.

The temperature in the algorithm gradually decreases according to a defined cooling schedule. At high temperatures, the system is more likely to accept worse solutions, which promotes exploration of the solution space. As the temperature decreases, the system becomes less likely to accept suboptimal solutions, thus encouraging convergence toward a better solution. Although Simulated Annealing can be computationally expensive and slower to converge compared to other methods, it is well-suited to this project due to the complexity of keyboard layout configurations.

3 Methodology

3.1 Keyboard Layout Representation

The keyboard layout is represented as a dictionary, where each key is associated with its corresponding (x, y) coordinate on the keyboard. Two standard layouts, QWERTY and Dvorak, are used as initial configurations for this study. Special keys such as `Ctrl`, `Alt`, `Space`, and `Shift` remain fixed during optimization. Only the alphabetic and numeric keys are subject to rearrangement.

Each key has an initial position on the home row, which serves as a reference point for calculating the total finger travel distance. The goal is to minimize the distance traveled from the home row while typing.

3.2 Distance Calculation

The total finger travel distance is calculated using the Euclidean distance formula. The sum of these distances across the entire text gives the total finger travel distance.

Additionally, after any key swaps, if either of the swapped keys is located in the home row, the starting position for all keys corresponding to that swapped key is updated accordingly. This ensures the correct calculation of distances post-optimization.

3.3 Text Analysis and Optimization

The text chosen for analysis is parsed using the `analyze_text_and_calculate_distance` function, which computes the total finger travel distance for the initial layout and records the frequency of key presses. A heatmap is then generated to visualize the frequency distribution across the keyboard. This provides a baseline for comparison with optimized layouts.

Once the initial analysis is completed, the Simulated Annealing algorithm is applied. The key steps of the optimization process are:

1. **Initialization:** The algorithm starts with an initial keyboard layout (either QWERTY or Dvorak).
2. **Neighbour Generation:** A neighboring layout is generated by swapping two randomly chosen keys.
3. **Cost Calculation:** The total finger travel distance for the new layout is calculated using the given text.
4. **Acceptance Probability:** The new layout is accepted with a probability that depends on the improvement in travel distance and the current temperature. Worse layouts are accepted with a probability proportional to the temperature.
5. **Cooling Schedule:** The temperature is gradually reduced, decreasing the likelihood of accepting suboptimal solutions as iterations progress.

This process is repeated over several iterations, allowing the algorithm to explore and refine the layout until convergence is achieved. The key parameters used for Simulated Annealing are:

- `initial_temp`: 100 (starting temperature)
- `cooling_rate`: 0.999 (rate at which the temperature decreases)
- `num_iterations`: 8000 (number of iterations per run)
- `num_runs`: 5 (number of independent optimization runs)

By running the optimization multiple times, we aim to identify the best possible layout that minimizes finger travel distance.

3.4 Layout Swapping Mechanism

The function `get_neighbour` is responsible for generating neighboring layouts by swapping two randomly selected keys. Special keys such as `Ctrl`, `Alt`, `Shift`, and `Space` remain fixed during optimization. This ensures that the function only swaps relevant keys and maintains a valid keyboard layout.

3.5 Visualization

Heatmaps are generated to visualize key press frequencies for both the initial and optimized layouts. These heatmaps provide insight into which keys are used most frequently and help to assess the impact of optimization. Additionally, a plot of the total finger travel distance over iterations is created to illustrate the convergence of the Simulated Annealing algorithm.

4 Results and Analysis

The results of the optimization process are compared across the two initial keyboard layouts: QWERTY and Dvorak. The optimized finger travel distance for the QWERTY layout was found to be approximately equal to the total finger travel distance for the Dvorak layout when typing the same text. This result highlights the inherent efficiency of the Dvorak layout.

The Dvorak layout is designed to place the most commonly used letters on the home row, where fingers naturally rest. This minimizes finger movement and enhances typing speed and comfort. The heatmap and cost plots show a significant reduction in total finger travel distance after optimization for both layouts, confirming the effectiveness of the optimization approach.

5 Conclusion

This project successfully demonstrates the use of Simulated Annealing to optimize keyboard layouts, significantly reducing finger travel distance during typing. The QWERTY layout was improved to approach the efficiency of the Dvorak layout, showcasing the potential benefits of optimization. Visualizations of key press frequencies and the cost of distance over iterations provide valuable insights into the optimization process and the resulting layouts.